

Minjae Seo

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EDUCATION

University of Chicago, Harris School of Public Policy

June 2027

Master of Arts in Public Policy, Certificate in Research Methods

GPA: 3.7/4.0

Coursework: Applied Econometrics I–III (PhD), Advanced Microeconomics for Policy Analysis I–II (PhD), Game Theory (PhD), Analysis in $\mathbb{R}^n$, Research Methodology and Writing

University of California, Berkeley

Dec 2024

Bachelor of Arts in Statistics

GPA: 3.85/4.0

Coursework: Modern Statistical Prediction & Machine Learning, Principles & Techniques of Data Science, Econometrics, Causal Inference, Optimization Models in Engineering, Game Theory

TECHNICAL SKILLS

Languages: Python, SQL, R, Stata, C, MATLAB, SAS

Toolkit: pandas, NumPy, scikit-learn, matplotlib, fixest, rdrobust, dplyr, ggplot2, Git, Quarto, LaTeX, AWS, WRDS Cloud, Tableau, Power BI, React, ETL

EXPERIENCE

Research Analyst Intern, World Bank, Washington, DC

Jun 2026 – Aug 2026

- Analyzed two nationwide multi-arm A/B experiments for Pakistan’s Federal Board of Revenue (88,229 companies across 8 message arms; 102,130 engineers across 6), where messaging raised filing by 2.06pp and 3.96pp but over 90% of the corporate filings it caused were empty.
- Established that only the engineer campaign moved revenue (+1.3pp positive liability, +PKR 455 per taxpayer), redirecting program reporting from filing counts to declared liability.
- Produced a 17-artifact reproducible R pipeline across five commissioned analyses, resolving 27 reviewer points and pulling a headline +81% figure that conditioned on a post-treatment outcome.
- Built the Lumina evidence dashboard on 30 verified youth-employment RCTs, rejecting one of 18 supplied references as not an RCT.

Research Assistant, Booth School of Business, University of Chicago

Jan 2026 – Jun 2026

- Delivered a 2.38B-row (4.90 TiB) U.S. job-postings dataset on WRDS Cloud using chunked state-by-year SQL with checkpoint/resume, so the extraction runs unattended and restarts from the last completed chunk.
- Consolidated 899 Excel sources (448,119 citation rows, 1920–2000) into a ranked 14,500-journal database, using fuzzy name matching, three-pass QC, and a parallel Python build so the ranking reconciles across two independent implementations.
- Mapped the library’s access coverage by classifying all 14,500 journals across four digital platforms.

AI Intern, YCX, Remote

Mar 2025 – May 2025

- Shipped an AI know-your-business compliance system to production, automating regulatory due-diligence review.
- Grew the companies the system can screen by 40%, to 300+, with a modular pipeline of LLM agents (Groq API) and targeted web scraping.

Quantitative Research Intern, Global Key Advisors

Jul 2024 – Mar 2025

- Engineered reusable Python extraction and ROI-tracking pipelines that fed the team’s downstream strategy backtests, merging SEC Form 4, CRSP, and Refinitiv across the Fama–French 48 industries.
- Tested whether insider-disclosure rules leave exploitable alpha using event-time abnormal returns, benchmarked against Fama–French factors.

PROJECTS

Community-College Transfer Earnings Gap (R, NLSY97 microdata)

Sep 2025 – May 2026

- Measured the earnings gap between community-college transfer students and four-year entrants: \$7,770 less per year ($p = 0.048$, $N = 2,306$).
- Confirmed the penalty across five estimators (OLS, propensity matching, IPW, AIPW), which bracketed it between \$7,770 and \$5,189, and matched a published administrative-data benchmark.
- Showed a widely cited explanation was a statistical artifact: holding the comparison group fixed shrank it from 27.5% to 4.5%.

Retail Credit Default Prediction (Python, scikit-learn)

Mar 2026 – Apr 2026

- Trained a default-risk model on a loan book with a 5.4% default rate and 100+ correlated features, using L1 selection, SMOTE, and cost-sensitive learning, and scoring on F1/AUCPR so the metric tracks caught defaults rather than accuracy.
- Wrote a concept-drift detection and retraining policy so post-deployment decay is caught on a schedule, with notebooks published.

AI Exposure and Labor Demand (Python, Revelio Labs + Compustat)

Mar 2026 – Jun 2026

- Assembled a 14.5M-cell panel linking Revelio Labs job postings to Compustat financials (2018–2025), which supported a triple-difference across occupation AI exposure, firm size, and the pre/post-ChatGPT split.
- Estimated sharp declines in AI-exposed postings at small (-0.211 , SE 0.033) and large (-0.146 , SE 0.026) firms, but reported the gap as a null ($+0.065$, $p = 0.14$).

TEACHING & LEADERSHIP

Graduate Teaching Assistant, Mathematical Foundations of ML, University of Chicago

Sep 2025 – Dec 2025

- Taught weekly recitations for 20+ master’s students, translating SVD, Gram–Schmidt, and least squares into Python/NumPy demos.
- Answered 50+ technical questions on Ed Discussion.

Head of Research, Undergraduate Economics Association, UC Berkeley

Jan 2024 – Mar 2025

- Led the research program of a 30+ member organization, running biweekly workshops in regression (OLS, IV, DiD, RDD) with R/Stata.
- Mentored members through semester-long projects, from identification strategy to written deliverables.

Competitor, Econometrics Game, University of Chicago

Mar 2024

- Selected as one of four (two LSE, one UC Berkeley) to produce a complete empirical study in 48 hours.
- Presented the team’s final analysis to a judging panel including a Nobel Laureate in Economics.